

Does fictionalism offer an adequate philosophy of mathematics? Is it able to accommodate phenomena such as mathematical constraint and mathematical discovery? Why, or why not?

In this paper, I will argue that fictionalism does offer an adequate philosophy for mathematics that accommodates various phenomena related to mathematics. I will first look at how the fictionalist theory impacts the validity of our definition of mathematics and then consider how this impact effects the viability of fictionalism as an adequate philosophy of mathematics.

To begin, we must first define fictionalism, both generally and as it pertains to mathematics. Fictionalism is defined as “about a region of discourse, [...] the view that claims made within that discourse are not best seen as aiming at literal truth but are better regarded as a sort of ‘fiction’”¹. Specifically, with respect to mathematics, it is the nominalist view that rejects mathematical Platonism. By a Platonist's view, when we talk about and carry out math, there exist abstract objects that represent mathematical quantities, thereby validating math. For example, the sentence “Mars is red” is true in virtue of the existence of real object, Mars, that has the property of being red. Likewise, a Platonist believes that saying “3 is prime” is true in virtue of the existence of an abstract object, 3, that has the property of being prime. On that other hand, the fictionalist view is that since these abstract objects do not actually exist, our mathematical sentences are false. Going back to Kant, there is no “truth-maker” object in the “real world” in virtue of which mathematical sentences are true. In a sense, they are as meaningless as saying something such as “Gandalf is a wizard”.

It is important to note, however, that fictionalists do not believe that this implies that the entire body of mathematics is false. Rather, they believe in indexing the truth value of a predicate on the “set of fiction” upon which the predicate applies. In the case of mathematics then, any sentence such as “ $2+2=4$ ” is true only within the “set of fiction” we know as mathematics but is false in the real world. Thus, under fictionalism, the real-world invalidity of a predicate does not in any way imply the invalidity of the set of fiction upon which the predicate acts.

So far, we've confirmed the fact that the fictionalist view does not invalidate mathematics as a whole but simply considers its predicates to be false in the real world. Now we consider if it is even possible to construct valid mathematics without reliance on objects, i.e. a nominalist theory of mathematics. The primary argument against mathematical nominalism is the Quine-Putnam Indispensability Argument³, which is as follows:

(P1) We ought to have ontological commitment to all and only the entities that are indispensable to our best scientific theories.

(P2) Mathematical entities are indispensable to our best scientific theories.

(C) We ought to have ontological commitment to mathematical entities.

On the other hand, Rosen and Burgess provide us with a strong argument for nominalism, which could be considered as, in their own words, the “dispensability argument”.² Its quite long so I won't include it here in its entirety but it states that since mathematics is essentially a scientific theory, it can be replaced by constructing another valid theory that achieves the same result but with “weaker substantive commitments.” Indeed, it is possible to construct nominalist alternatives to current mathematical theories.

This was first done by Hartry Field in his 1980 work, *Science Without Numbers*, who provided a detailed and “complete axiomatization of Newtonian mechanics with no reference to numbers or functions at all”, as well as general ideas on how to extend his theories into other fields, particularly Maxwell's theory of electromagnetism.^{4,5} Yablo also provides a similar example in his book, *Myth of Seven*, wherein he shows the derivation of important bases of mathematics using only quantifiers and relations between propositions. This is all done without using numbers themselves (though he does provide the definition of finite and infinite numbers).

Now that we have partly constructed a nominalist theory of math, or at least shown the viability of such a construction, we must consider its validity. Fictionalism considers mathematical sentences to be fictional, almost akin to metaphors, and as such, we can evaluate their validity in a similar way. A metaphor is simply a specific type of model that we used linguistically to represent something else. For example, instead of saying “Time to leave”, one could say “Time to hit the road”. This linguistic model is far more accurate if one is travelling to a different destination rather than simply going to their next class. The validity of a model is given by its predicative capabilities; a better, i.e. more valid, model predicts its target more accurately.

Thus, under a fictionalist view, we can measure the predictive capabilities of mathematics on the real world. We can do this by substituting real world objects into mathematical theories to see if they still hold. Consider the “set of fiction” given by mathematics; here, all our mathematical sentences hold (ex: “ $2+3=5$ ”) and their components (ex: “2”, “3”, “5”) represent some abstract objects. An important distinction is that here, abstract object does not represent the same definition as that in the Platonic view, but object, regardless of the definition does not exist in the real world and so our fictionalist view is preserved. In order to translate this to the real world, we can replace these abstract objects in the “set of fiction” of mathematics with real world equivalents. For example, we can replace “2” and “3” with golf balls. If we were to add 2 golf balls into a bag and then add 3 more, the bag would contain 5 total balls, validating our theory. This is a very simple example, but this can be extended for various other aspects of mathematics. Using golf balls as a representation of numbers is like using props to support mathematics, per Yablo. He argues that mathematics is a make-believe game and we can use real world items as props. Since mathematics is the investigation of all possibilities, it makes sense that one of these possibilities can model the real world adequately.

However, with this example, we do encounter a limitation on the formalist view, namely that the above representation does not work for every single theory of mathematics. Consider, for example, n -dimensional geometry, with $n > 4$. For $n < 4$, we can create physical objects which can represent various mathematical structures and for $n=4$, we can use abstract art to attempt an adequate representation. For example, we can use a GIF to represent a tesseract, 4-dimensional cube. The field of geometry is simply one of many cases where we cannot directly validate mathematical theory by generating physical representations. Although we cannot generate physical representations of all mathematical theories, that does not mean that these theories will not still yield accurate models of the real world. Consider Chern-Simons forms⁶ for example: they are a mathematical description regarding certain properties in the field of topology and it would be basically impossible to create a physical representation of this theory. However, these forms have important implications in fields of quantum mechanics and string theory.

This brings us to the question of mathematical constraint. Since we cannot determinedly validate all our theorems using an analog in the physical world, we aren’t able to determine the accuracy of our model. As mentioned before, fictionalism considers mathematics to be true within a set of fiction. In this set of fiction, all of mathematics as we know it holds true under classical logic, namely the law of non-contradiction. Since our basics of mathematics can be validated, as mentioned above, all other aspects can be derived from this and we have guaranteed validity.

Another related limitation is one due to lack of understanding. When using a linguistic metaphor, if someone doesn’t understand, you can point them towards the literal meaning in order to help them understand. For example, the metaphor “rise and shine” is used in the morning in the sense of “wake up and start the day” and doesn’t literally mean “elevate and glow”. If someone doesn’t understand this metaphor, you can inform them that it originates from the fact that the sun rises and shines in the morning, and the sun is often used to represent the morning. However, in the case of mathematics, if someone has difficulty understanding a mathematical theory, especially more complicated ones, under fictionalism, there are no actual objects or literals that you could point them to aid their understanding. They must simply work through the mathematics in order to understand.

We’ve now shown that the validity of mathematics still holds under fictionalism, but we have yet to verify whether this is an adequate philosophy of mathematics. We already showed that it can accommodate mathematical constraint but now we will check accommodation of mathematical discovery. It is related to constraint and has a similar “proof”. As mentioned above, within the appropriate set of fiction, we can use logic, specifically the law of non-contradiction, as usual. Thus, any new mathematical discoveries still hold under fictionalism as long as they obey the rules of mathematical constraint. However, they are treated as before, being considered false in the real world and only true in the set of fiction of mathematics.

I believe that this does show that fictionalism is an adequate philosophy for mathematics but there remains the question of whether it has any benefits. Most people who do mathematics do not even consider the philosophy behind the math and of those who do, at first most people adopt a Platonist view. It is quite easy to grasp and provides a very sufficient philosophy for understanding mathematics. Thus, it is important to see whether fictionalism is a precise enough philosophy for mathematics that the average mathematician would adopt it. There do exist many anti-Platonist alternatives to fictionalism, but I will not cover them here and will only speak of the choice between fictionalism and Platonism as it relates to mathematics. I chose to do this due to their opposing natures with respect to nominalism. To see if one option is more favorable to us, we can apply Occam's Razor. Going from Platonism to fictionalism, we can remove the ontological dependence on abstract objects without loss of explanation, we are vindicated in doing so.

This leads us to believe that fictionalism is superior to Platonism; however, several proponents of fictionalism itself reject this theory, chief amongst whom are Rosen and Burgess. In their 1997 book, *A Subject with No Object*, they suggest that "issues of rigour and consistency cloud ontological debates in mathematics to such an extent that it is difficult to know whether economy of abstract ontology really is a virtue of scientific and mathematical theories".⁹ Further, Mark Balaguer, another fictionalist, argues that it is possible that fictionalism does not account for every single thing that Platonism might account for. Fictionalism is also not simpler than Platonism in all non-ontological respects; the latter "enables us to say that all our scientific theories are [...] largely true."¹⁰

In conclusion, fictionalism does indeed offer an adequate philosophy of mathematics. It provides us with the capacity for both constraints and discovery within mathematics, although it maintains that any new discoveries are still false. Further, there is not apparent superiority between choosing fictionalism and Platonism as neither provides an explicit benefit over the other. In fact, I believe another solution altogether might be more beneficial: mathematical inferentialism.⁷ It posits that "mathematical sentence can perform the function of facilitating deductive inferences from some concrete sentences to other concrete sentences". It does maintain that these sentences are "false", as in fictionalism.

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